Benchmarking Transformers for Deepfake Detection: Analysing Performance and Identifying Research Opportunities

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Abstract. Deepfake detection has become a critical area of research due to the increasing misuse of AI-generated media for spreading misinformation. Existing detection techniques, while effective, often struggle with generalization across diverse datasets. Unlike prior studies, which often focus on a single dataset or model, this research provides a holistic across three widely-used datasets Face Forensics++, Deepfake Detection Challenge (DFDC), and Celeb-DF. An evaluation of the traditional transformer to SOTA models sheds light on the baseline performance of transformers for deep-fake detection. The research demonstrates that while transformers, particularly those incorporating both spatial and temporal information, exhibit strong detection capabilities up to 92%, their performance varies significantly depending on the dataset. Based on the evaluations it was deduced that the baseline transformer's accuracy was at 66.2%, the Convolutional ViT model had 70.3%, the ADT model 82. 6%, Efficient ViT 76.2 % while the Pooling Transformer had 68.1% that is followed by the Spatiotemporal Transformer which achieved an accuracy of 86.1% accuracies. The evaluations done also revealed some significant shortcomings of the current transformer models including poor generalization performances even on high-quality deepfake samples and variability in the performances and stressed the necessity for the enhancement of better transformer structures.

1. INTRODUCTION

Recent years we have seen an increase in the development of the deep learning approaches which have brought revolution in many fields such as computer vision, natural language processing, audio and video generation. Of those innovations made by deep learning, Deepfake technology [1-2] has become famous due to the possibilities to create nearly realistic fake videos that can be used to spread fake news. Deepfake technology operates by training models on large datasets of real images and videos to generate fake media that mimics the appearance and mannerisms of the subject [4]. The ability to detect and mitigate the impact of Deepfakes is crucial for maintaining trust in digital media, safeguarding public figures from malicious attacks, and ensuring the integrity of information shared online [5]. Existing methods for Deepfake detection often rely on traditional machine learning and deep learning models, which, while effective to some extent, face challenges in generalization, scalability, and robustness which has led to the exploration of more advanced architectures, such as Transformer networks [6-7]. Main contribution of the work can be summarized as follows. This work offers a thorough comparative analysis of state-of-the-art transformer models specifically for Deepfake detection across three of the most widely used datasets FF++, DFDC, and Celeb-DF. Through the detailed analysis of various state-of-the-art (SOTA) transformer networks, this research uncovers specific gaps in the current literature, particularly concerning the generalization capabilities of transformer models for Deepfake detection task.

2. LITERATURE REVIEW

The first approach to deep fake videos detection was based on conventional machine learning methods including SVM [11] and decision trees [12] with manually engineered features [29-32]. These methods cantered on identifying disparities of pixilation [13], illumination, or other informative disparities intentionally incorporated into a deepfake during its creation. But were constrained by the features in their lack of cross data set and manipulate well generalizability. To overcome limitations of traditional methods, researchers have stared incorporating the deep learning models such as CNNs [14], [33-34] and RNNs [15] into the detection processes. In other words, LSTM model capable of analysing temporal features while XceptionNet model [16] is able to analyse spatial characteristics improving the detection rate depending on both frame level and sequence level. However, these methods are not very effective in detecting high quality realistic deepfakes that are created by the current state of the art GANs.As far as the architectural adaptations are concerned one of the native techniques is the convolutional vision transformer (CViT) [17]. More recently, ADT [18] addresses the shortcomings of traditional CNN-based methods by effectively capturing both global and local information. The Efficient ViT (Vision Transformer) [7] is a hybrid architecture designed to combine the efficiency of convolutional neural networks (CNNs) with the power of Vision Transformers. The Convolutional Vision Transformer (Convolutional ViT) [17] is a hybrid architecture that merges the strengths of Convolutional Neural Networks (CNNs) with those of ViTs to address the task of deepfake detection. These models have been tested on DFDC, FF++, WildDeepfake [19], Deeper Forensics [20] and Celeb-DF datasets and it has been seen that the proposed models outperform the traditional CNN based models [21-28]. However, there are still some issues associated with transformers, such as the amount of computational cost and resources that is needed for training these models. Thus, current work is focused on improving these models as well as real-time detection and implementation of such models in limited-resource environments.

3. TRANSFORMER ARCHITECTURE

The Traditional Transformer is a pioneering model introduced by Vaswani et al. in 2017 [6] in their paper. It revolutionized natural language processing (NLP) by relying entirely on self-attention mechanisms instead of recurrent or convolutional layers to process sequential data. The architecture consists of an encoder-decoder structure, where the Encoder processes the input sequence and generates a set of encoded representations and the decoder takes the encoder's output and, together with the target sequence, generates the final output. Anti-Deepfake Transformer (ADT). The Anti-Deepfake Transformer (ADT) [18] is a complex transformer-based model, proposed inherently for classification of deepfakes. The original spatial structure of the input image is retained by subdividing the image into overlapping patches. Specifically, ADT utilizes several trans-blocks where transformer layers are accumulated, the architecture applied multi-Head Self-Attention, and Multi-layer Perceptron. These layers first extract the patches in which the patterns of interest reside, and then transform them to capture features at long-range scales in the entire image. Attention Leading Module (ALM) aim at making the model pay attention to the highly discriminative areas of the image where deepfake artifacts are is more likely to be found. The architecture is built upon two primary components: used an EfficientNet as the feature extractor or extractor and a Vision Transformer as the encoder. The architecture is a lightweight and competitive CNN, which is in charge of feature extraction from face images. The features obtained by EfficientNet B0 are flattened, linearly projected to tokens and passed to Vision Transformer encoder. These tokens are then fed to the encoder with the use of a self-attention mechanism that incorporates global dependencies and patterns of the image. This enables the model to find fine and pervasive distortions which are not easily noticed when using basic convolutional methods alone. A classification head is appended after the transformer encoder, which uses the CLS (classification) token to output a binary decision (real or fake). The Efficient ViT model was selected for evaluation because it provides a highly efficient solution to the computational challenges posed by deepfake detection by leveraging EfficientNet for feature extraction. Moreover, the integration of a Vision Transformer allows the model to handle complex patterns in video frames, which are crucial for detecting sophisticated deepfakes.

Convolutional ViT model is also built to incorporate insight from CNNs aspects of spatial locality and transformers global context modelling paradigm. The obtained output feature maps are then divided into several patches. These patches are flattened and inserted into a higher dimensional space for use as tokens for the Vision Transformer. The embedded patches are passed to the transformer encoder and are subjected to multi-head self-attention to understand various patches' connexion. This makes it possible to account for feature dependencies and other intricate features that may cross the entire image region but are hard to capture by the traditional model. Furthermore, the output of transformer encoder passes through a classification head that classify whether the input image or a sequence of images is real or fake. The prediction made by this classification head usually employs the CLS token from the transformer encoder. Thus, the Convolutional ViT model was evaluated because of its benefits of the combination of the convolutional neural networks and the transformers. The pooling transformer model uses Hybrid of CNN and transformer blocks to learn hierarchical features so that it can be highly effective against hyper realistic deepfake videos. The first step in pooling transformer is to have a Convolutional Feature Extraction where an input image passes through several convolutional layers to extract features at the local level. The features extracted are passed to Pooling Transformer and this transformer is further divided into three stages and each of these stages is separated by a pooling layer. The pooling layers help in enhancing the feature maps where they decrease the spatial dimensions of the maps while at the same time doubling the number of channels to direct the model's attention to only important features. This is then supplanted by a forced re-attention mechanism that helps to restore diversity in attention maps across transformer layers to prevent the model from forgetting the globally important information as it progresses deeper into its depth. But for the improvement in the learning capability of the transformer, a re-attention matrix is used which consists of the heads of the attention and this portion also sees to it that the given heads do not cater same attention. The Spatiotemporal Transformer involves both CNNs and Vision Transformer (ViT) for improving deep fake detection. The model begins with an input video which contains one or more individuals; from the video. The cropped face images are then fed to a CNN, namely, the Xception Network in-order to learn the high-level feature maps. Each of the extracted feature maps are then flattened and input into a ViT encoder module which exploits the self-attention mechanism. The transformer encoder outputs are then given to a MLP classification head for fake/real discrimination. The Spatiotemporal Transformer was chosen for the testing due to the fact that it designed to model both spatial and temporal information necessary for identifying deepfakes.

4. EXPERIMENTAL ANALYSIS AND RESULTS

The experiments for deepfake detection were conducted using the PyTorch framework, which provides robust tools for developing and testing deep learning models. The training and evaluation processes were performed on Google Colab, leveraging the computational power necessary for training transformer-based architectures efficiently. The input images, extracted from video frames, were resized to a resolution of 256x256 pixels to standardize the input dimensions and optimize computational resources. Data augmentation techniques, such as random cropping and horizontal flipping, were applied during training to improve the model's robustness and generalization capabilities.

Overview of Datasets Used

For the experiments FF++, DFDC, and Celeb-DF datasets are used to evaluate the transformer models, these datasets are quite common in the literature for deepfake research and have seen significant contributors as shown in Fig. 1.

FIGURE 1. Normal and Deepfake samples taken from the FF++ dataset [Source: Author].

FF++ is among the most extensive compared to the other datasets. FF++ has a large database of face swapped video clips produced with help of different modern face swap techniques. The DFDC was designed as an effort of the Facebook AI Research division along with repositories and AI-proliferating organizations across the world to help encourage the rise of powerful deepfake-detecting algorithms. The DFDC dataset encompasses more than 100K video clips of different people targeting different age, ethnic, and gender distribution. Another quite popular dataset in the sphere of Deepfake detection is Celeb-DF, which contains a great quantity of high quality and quite realistic Deepfake videos. The dataset includes 590 real videos from YouTube, where 59 celebrities of varying ethnic origin.

Training Loss

In Fig. 2, MSE loss plots of transformer models have been depicted at 100 epochs where these models have been trained on FF++ DFDC and Celeb-DF datasets.

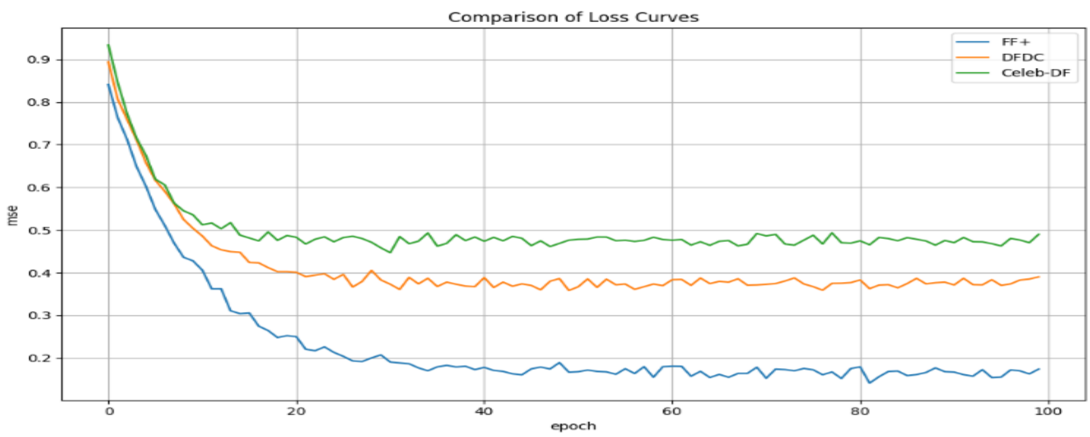

FIGURE 2. Training loss of the traditional transformer model over the epochs [Source: Author].

The transformer model yields most significant improvement in MSE loss in FF++ around 0.1665, this may be because the transformer model is extremely efficient in learning and applying knowledge from this dataset, and it could equally be as a result of the improved organization and simplicity of manipulations in FF++ as compared to the other sets of data. Contrary to other tested datasets, the reduction of MSE in the DFDC dataset is rather slow and stabilizes at about 0.374. This means that the model has to handle more variation and some level of complexity in the DFDC data set that may contain a lot of manipulations as well as high levels of noise or artifacts lowering the achievable loss. Celeb-DF is least able to decrease the MSE and it settled for around 0.473. From this, it is obvious that the model finds it difficult to detect deep fakes in this dataset because deepfakes are most realistic.

1. SOTA Accuracy

The Table 1 presents the SOTA performance of various transformer-based models on three datasets: Four datasets which are FF++, DFDC, and Celeb-DF were used. The result of each model is expressed by accuracy (%) which is one of the main and valuable criteria to compare deepfake detection models. Convolutional ViT scores relatively well on FF++ (67%) and DFDC (73. 6%) while it does not have results on Celeb-DF, despite the fact this method might not have been tested on all the datasets or it is less efficient for high-quality deepfakes such as those in Celeb-DF. ADT has 76.2% on DFDC and it is 89. 0% on Celeb-DF – arrangements which prove how capable it is in dealing with the complex shifts common to these datasets. Traditional transformer achieves high accuracy in detecting FF++ (83.2%) while its performances in detecting more challenging or realistic deepfakes in DFDC (62.6%) and Celeb-DF (52.7%) are considerably poor. Pooling Transformer outperforms Traditional Transformer with respect to FF++ with 87.4% and slightly better performance in cases of both DFDC and Celeb-DF with 62.7% and respectively 54.3%. These results indicate that the computation is a great way to enhance generalization, however, it has its drawback in difficult datasets. Finally, it is clear that the spatiotemporal transformer is the best model among the high performers across all the datasets with an average value of 85. 60% and with outstanding performance in FF++ with a score of 92. 6%, DFDC with a score of 89. 7% and Celeb-DF with a score of 76. This means that the inclusion of both spatial and temporal information greatly improves the model’s capability of identification of deepfakes at various resolutions and levels of processing.

TABLE 1. Side by side evaluation of the accuracy of the various transformers over FF++, DFDC and Celeb-DF datasets [Source: Author].

Method	**FF++**	**DFDC**	**Celeb-DF**
Traditional Transformer	83.2%	62.6%	52.7%
Convolutional ViT	67.0%	73.6%	-
ADT	-	76.2%	89.0%
Efficient ViT	76.2%	-	-
Pooling Transformer	87.4%	62.7%	54.3%
Spatiotemporal Transformer	92.6%	89.7%	76.0%

2. Evaluation Summary and discussions

While transformer models have shown great promise in deepfake detection, there are still several challenges exist that need to be addressed for deep-fake detection. One major challenge is the lack of generalization across different datasets and manipulation techniques. For instance, traditional Transformer had average accuracy of 66.2%, Convolutional ViT had 70.3%, the ADT model 82.6%, Efficient ViT 76.2%, Pooling Transformer having 68.1% and finally the Spatiotemporal Transformer boasting 86.1% accuracies. Most transformer-based approaches require high-performance GPUs and substantial computational resources, making them impractical for deployment in real-time applications or on edge devices.

5. CONCLUSIONS AND FUTURE SCOPE

This work has provided a comprehensive evaluation of transformer-based models for deepfake detection across three prominent datasets FF++, DFDC, and Celeb-DF. The study evaluated various transformer models compared to a baseline transformer for Deepfake detection in terms of accuracy. Findings demonstrate that while SOTA transformer models, exhibited good accuracies and robustness. However, their performance varies significantly depending on the dataset. These limitations suggest a need for further research into optimizing transformers for deep-fake detection and improving their generalization capabilities. For future works major area that requires further exploration is the integration of spatio-temporal features into transformer models. Although some studies have attempted to incorporate temporal features through the use of LSTM or GRU layers, there is still a need for more effective methods to capture the dynamic nature of video data in deepfake detection. Future work should focus on developing such architectures and integration of contain visual artefacts as with better attention mechanisms to ensure that transformer models can be effectively deployed for the real-world scenarios.

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